Product Assurance Plan for Manufacturing Procedure

Process 1	Preparation of product assurance plan for manufacturing	Personnel resource	No designation
		Material resource	No designation

INPUT

The suppliers shall develop a product assurance plan for manufacturing to control the progress of the activities specified in the stage management with taking into account the following items, but are not limited to the following:

a) Linkage to each stage (see Figure 1 and Table 1)
b) Control items and achievement targets (see Table 2)

<u>In addition, the suppliers must comply with any other directions given by Honda regarding control items and achievement targets in the product assurance plan for manufacturing.</u>

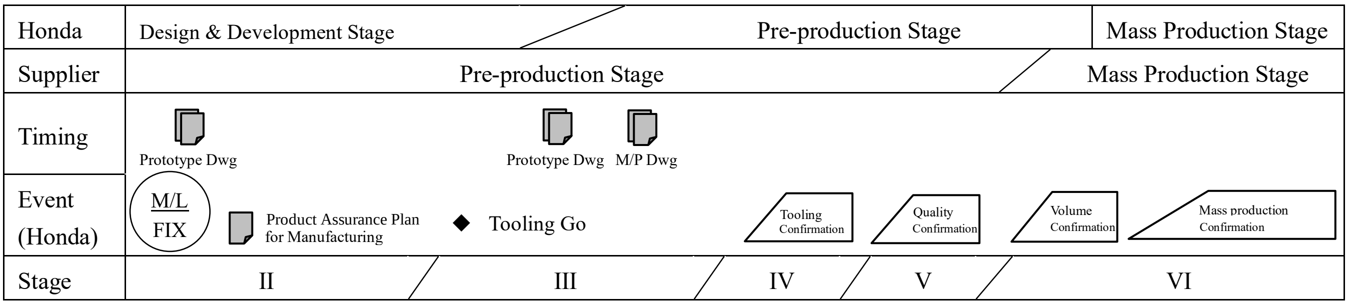

Figure 1 Stage image

Table 1 Overview of each stage

Stage	Definition
Stage I	Obtain product requirements from Honda (Stage I will be omitted hereafter since this stage consists of activities prior to the Pre-production)
Stage II	Formulate a product assurance plan for manufacturing
Stage III	Promote activities required based on the product assurance plan for manufacturing
Stage IV	Verify quality maturation status
Stage V	Confirm the prospect of transition to mass production and declare completion of production preparation
Stage VI	Mass production

Note

The suppliers may coordinate with Honda on a basis of product with respect to the contents of Table 2.

Table 2 Control items and achievement targets

No.	Control Item	Stage/Category Outline	Pre-Production Stage				Production Stage
			Stage II	Stage III	Stage IV	Stage V	Stage VI
1	Supplier's production plan.	A supplier's production plan for the pre-production stage drawn up in line with Honda's production plan.	• Draw up a plan	• Reflect delivery and quantity to the plan.			• Address concerns from the previous stage. • Production and delivery in line with Honda's production plan. • Practice quality control in line with "SQM 5.1 Initial Quality Control in Mass Production" and other requirements of the SQM.
2	Tooling Plan	Draw up plan for designing, manufacturing and application of dies and molds (press, plastic and casting and forging) adjusted to the required level. If there are more than two dies or molds, draw up a plan per die or mold.	• Decide quantity of dies and molds. • Decide manufacturers. • Decide plans per die or mold.	• Manage planned and actual outputs. • Measures against the differences between planned and actual outputs.	• Trial with final and permanent model dies or molds. • Product evaluation.	• Finish up final model dies or molds (hardening process, etc.) sintering • Confirm by continues production	
3	Production facility setup	Installation plan for facilities (new, refurbished, etc.) necessary to secure required quality (production capacity, included).	• Decide production facility concept. • Decide schedule for facility setup.	• Manage planned and actual outputs. • Measures against the differences between planned and actual outputs.	• Perform line trials with permanent facilities. • Product evaluation. • Process capability verification.	• Confirm by continues production • Check quantity. • Verify process capability.	
4	Jig/testing device/inspection equipment setup (dimensions and accuracy).	Plan to secure needed jigs/testing devices/ inspection equipment to guarantee required quality.	• Schedule a date to set up jigs/testing devices/inspection equipment and complete process design.	• Manage planned and actual outputs. • Measures against the differences between planned and actual outputs.	• Confirm the accuracy of jigs/testing devises/inspection equipment.	• Check the operation state of inspection equipment in the line during continuous production test runs	
5	Validity testing	A plan that will be used to confirm required part characteristics, based on roles and responsibilities agreed with Honda. Carry out tests on specifications specified in the applicable drawing (specifications included).	• Concerns on required quality and measures against them • Draw up a test plan.		• Commence variation testing. • Progress performance management against the plan.	• Complete variation testing.	
6	QA device validity testing (function)	A plan used to validate QA device.	• Decide QA device concept. • QA device setup plan	• Manage planned and actual outputs.	• Verify QA device.	• Check the operation state of QA devices in the line during continuous production test runs	

7	Production capability	Plan to determine the number of man-hours and manpower needed for the planned volume of vehicle production to ensure delivery time and quantity.	• Set the maximum production capacity and decide schedule for procurement of needed resources.	• Manage planned and actual outputs. • Measures against the differences between planned and actual outputs.	• Perform line trials with permanent facilities.	• Verify continuous production • Check quantity.	
8	Process proficiency of operators.	Develop required control documents and draw up plans for improving operators' proficiency in the line.	• Draw up training plans.	• Commence training with actual parts. • Draw up control documents.	• Commence training with required line takt time.	• Complete operators' proficiency requirements.	

No.	Control Item	Outline \ Stage/Category	Pre-Production Stage				Production Stage
			Stage II	Stage III	Stage IV	Stage V	Stage VI
9	Color matching verification	A verification plan to reproduce the required color.	• Know about specified parts and their color. • Draw up verification plans.	• Manage planned and actual outputs. • Select paint.	• Agree the final master sample	• Agree the master sample for mass production products.	• Address concerns from the previous stage. • Production and delivery in line with Honda's production plan. • Practice quality control in line with "SQM 5.1 Initial Quality Control in Mass Production" and other requirements of the SQM.
10	Grain and gross matching verification.	Plan to reproduce the required appearance (grain and gross).	• Understand the scope of subject parts. • Draw up a plan.	• Manage planned and actual outputs.	• Agree the final master sample. (depth or trends of impression of grain pattern, etc.).	• Agree the master sample for mass production products	
11	Illumination master	Plan to ensure the illumination specification comply with the master sample.	• Understand the scope of subject parts. • Draw up a plan.	• Manage planned and actual outputs.	• Agree the final master sample.	• Agree the master sample for mass production products	
12	Feeling master	Plan to ensure feeling complies with the master sample.	• Understand the scope of subject parts. • Draw up a plan.	• Manage planned and actual outputs.	• Agree the final master sample.	• Agree the master sample for mass production products	
13	Control items at each manufacturing process	Plan to conduct process FMEA, etc. and determine items requiring control in a process.	• Determine possible causes by conducting process FMEA.	• Decide on control items for the manufacturing process.	• Process improvement.	• Process improvement	
14	Lot No. Display Detail	Plan to define lot control numbers to display on the parts, hand in "Lot No. Display Details" to Honda, and to carry out lot identification.	• Understand the scope of subject parts. • Decide the Lot No. Display Details. • Process set-up plan for lot identification.	• Manage planned and actual outputs.	• Check devices for display of part lot.	• Check continuous lot number display in the line.	
15	Process quality control table (Process flow charts/control plan)	An action plan for the process quality control table.	• Decide process sequence (process flow charts) • Formulate the process quality control table.		• Verify processes in line with the process quality control table.		

No.	Control Item	Outline \ Stage/Category	Pre-Production Stage				Production Stage
			Stage II	Stage III	Stage IV	Stage V	Stage VI
16	Approval mark	Application for part approval and approval mark display.	• Confirm regulatory requirements. • Understand the scope of subject parts. • Draw up a plan for gaining approval.	• Manage planned and actual outputs. • Agree on approval mark identification methods. • Part approval application.	• Evaluate products (conformity with drawings).	• Gain part approval. • Display an approval mark on the parts. • Submit an approval certificate and test reports to Honda.	• Address concerns from the previous stage. • Production and delivery in line with Honda's production plan. • Practice quality control in line with "SQM 5.1 Initial Quality Control in Mass Production" and other requirements of the SQM.
17	Material identification	Material identifiation for the recycling of materials of parts.	• Understand the scope of subject parts.	• Agree identification methods	• Evaluate products (conformity with drawings including specifications).		
18	Management of compliance with laws and regulations.	Plan to ensure compliance of products with laws and regulations.	• Confirm restrictions. • Understand the scope of subject parts/materials. • Draw up a plan to ensure compliance.	• Confirm conformity of purchased materials.	• Carry out inspection to ensure compliance with statutory and regulatory requirements.	• Present or submit inspection results.	
19	Mass Production Transition Declaration	Confirm transition of production to mass production.				• Mass Production Transition Declaration	